DocQuorum

Agentic document extraction with open models:
the pipeline on real benchmark documents

October 2026

Every example is a real benchmark document and a stored pipeline output; no model was called to make these slides.

Outline

1. Task and pipeline
2. Walkthrough on real documents
   inputs, routing, parsing, readers, deterministic checks, selection by quorum, output
3. Results on six benchmarks; zoom re-reading
4. Summary

Setting

Open-weight models only: Qwen3.8-27B (FP8) for reading and MinerU2.5 for parsing scans, served on a single GPU. Each benchmark is scored with its own official scorer.

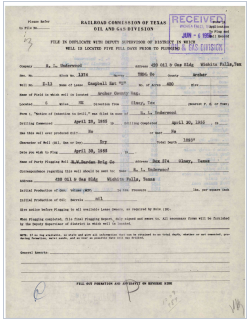
Problem statement

Task

Input: a document D (PDF or image), a JSON Schema S or a JSON template, and optionally a task description T.

Output: one JSON object Y that conforms to S, every value supported by D, absent values `null`.

Optional task (real example, receipt schema):
"A single point-of-sale purchase RECEIPT (retail or restaurant). DOC-TYPE / SCOPE RULE: if one file bundles MULTIPLE . . . "


RAILROAD COMMISSION OF TEXAS
OIL AND GAS DIVISION

RECEIVED

FILE IN DUPLICATE WITH DISTRICT SUPERVISOR OF DISTRICT IN WHICH WELL IS LOCATED FIVE FULL DAYS PRIOR TO PLUGGING

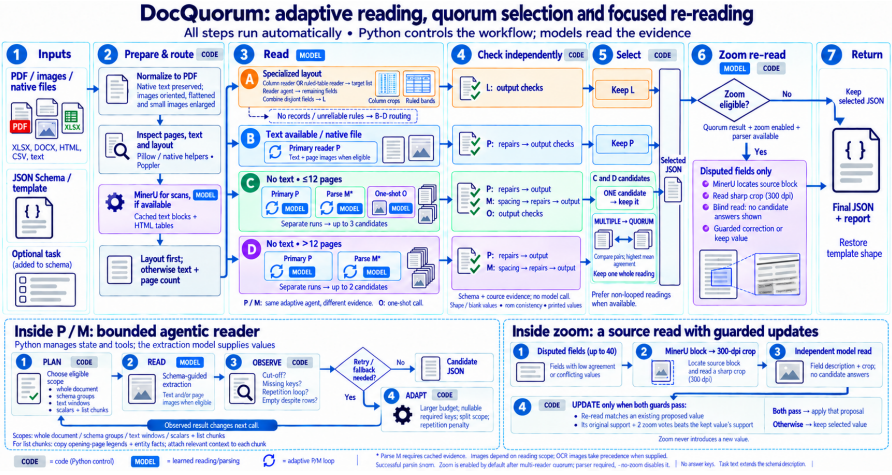
Scanned 1955 well-plugging application (Railroad Commission of Texas), 2 pages, no text layer.

JSON Schema (77 fields), excerpt

```
"p1_company":  {
  "anyOf":  [{"type":  "string"},
             {"type":  "null"}],
  "description":  "The 'Company' field
    in the identification block on
    page 1 ...  Extract verbatim as
    printed; null if blank.  Do NOT
    confuse with ..."
},
"p1_county":  {...},
"p1_total_depth":  {...},
...  74 further fields
```

ExtractBench short: 2A-9-160298_109929 (schema shortened); task text from the schema of hingham_lowes_receipt.

Pipeline overview



Pipeline figure (generated with Codex), describing the single-document entry point `docquorum.extract`; the lower panels show the bounded reader loop and guarded zoom updates (v1.3; v1.4 adds no step).

Step 2: the available evidence decides the route



A Layout reader

Unclaimed-property register
59 pages, 15,280 char./page



B Reader agent: text + images

County purchase order
1 page, 2,253 char./page



B Reader agent: text + images

W-14 well form
1 page, 252 char./page (OCR layer)



C Three readings, then quorum

1955 plugging application
2 pages, no embedded text



D Reader agent on the parse

Tax return
25 pages, no embedded text

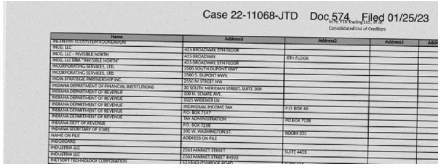
Routing rule

Layout first: pages printed as equal-width columns of records, or a ruled table on a scan, go to a layout reader A. Otherwise a PDF with embedded text (at least 50 characters per page; a digital PDF or an OCR layer) is read once B. A scan without text is parsed by MinerU and read three ways if it has at most 12 pages C, or read through its parse D.

ExtractBench documents; embedded characters per page measured with pdftotext.

Step 2: MinerU returns a scanned table as cells

Scanned page (detail)



MinerU output (raw HTML, abbreviated)

```
<tr><td>Name</td><td>Address1</td><td>Address2...<td>INCENTIVE
ECOSYSTEM FOUNDATION</td><td...<td>INCO, LLC</td><td>425
BROADWAY, 5TH FL...
```

The same rows as table cells (65 rows on this page)

Name	Address1	City	State	PostalCode
INCENTIVE ECOSYSTEM FOUNDATION				
INCO, LLC	425 BROADWAY, 5TH FLOOR			
INO, LLC - INAVAILABLE NORTH	425 BROADWAY	NEW YORK	NY	10013
INO, LLC DBA "INAVAILABLE NORTH"	425 BROADWAY, 5TH FLOOR	NEW YORK	NY	10013-2599
INCORPORATING SERVICES, LTD.	3500 SOUTH COOPONT HWY	NEW YORK	NY	10013

Columns stay aligned: an empty cell stays empty rather than pulling the next value into its place. Recognition errors remain, so the reader receives the page image together with the parse.

ExtractBench long: real_ftx_full_corrupted, page 47; cached MinerU2.5 parse.

Step 3, route A: the column reader



COLLINS RICHARD
EATON BETHINA
RT 1 BOX 220
JOHNSON JOHNEY B
C/O MAE JEAN JOHNSON,
RT 1 BOX 230
SHERRILL IVANNA
RT 1 BOX 1142

PROCTOR

BLEVINS JAMES DEE
RR2 BOX 556
SANDRA COTTONGAME
67000 S 367 RD

STILWELL

ADAIR TERI
RT 2 BOX 1155
BEASLEY ALAN C
470241 E 880 RD
BRD ALFRED JOSEPH
RT 2 BOX 271A
BREWER VENA MAUD
301 S CHEROKEE ST
BRIDGES INEZ
77150 HIGHWAY 59
BROWN CLIFFORD LEE
RT 4 BOX 2570
BROWN LISA
RT 3 BOX 2119
CRAWFORD BRADLEY
212 6TH PL
CRITTENDEN JESSIE
RT 6 BOX 1760
EUBANKS JR SHELBY
RT 4 BOX 1474
FOURKILLER HENRY
471958 E 810 RD

Records returned for column 1 (page 2)

name	locale
COLLINS RICHARD	null
EATON BETHINA	RT 1 BOX 220
JOHNSON JOHNEY B	C/O MAE JEAN JOHNSON, RT 1 BOX 230
SHERRILL IVANNA	RT 1 BOX 1142

7 columns detected on the page

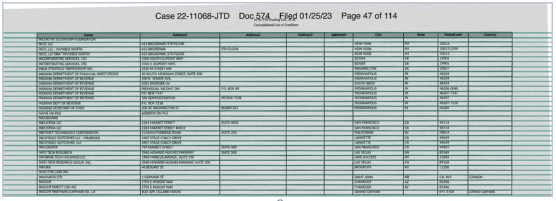
59 pages

26,675 records in the final output

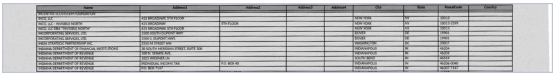
The trigger is page geometry only: at least three text columns of near-equal width. Each column is cut out and read top to bottom in one call; a record continued from the previous column is joined to it.

ExtractBench long: real_oklahoma_unclaimed_2024, page 2. Column boxes drawn with the pipeline's own geometry code; records from the stored final output.

Step 3, route A: the ruled-table reader



Printed rules detected after deskewing by 1.15° (green).



One call: the header row stacked on a band of 12 rows; the reply must contain exactly 12 entries.

ExtractBench long: real_ftx_full_corrupted, page 47. Rows from the stored final output.

67

rows counted from the rules

67

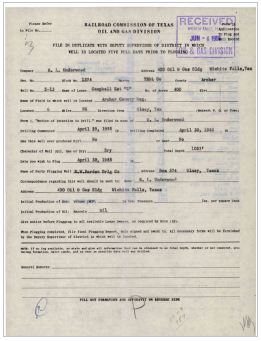
rows returned by the reader

Bands on page 47: $12 + 12 + 12 + 12 + 12 + 7$.

name	address_1	city	state
INCENTIVE ECOSYSTEM FOUNDATION	null	null	null
INCO, LLC	425 BROADWAY, 5TH FLOOR	NEW YORK	NY
INCO, LLC - INVISIBLE NORTH	425 BROADWAY	NEW YORK	NY

Fixing the number of entries per band prevents repeated identical rows from multiplying or disappearing.

Step 3, route C: three independent readings of one scan



field	P image reading	M parse reading	O one-shot
p1_company	R. L. Underwood	R. L. Underwood	null
p1_county	Archer	Archer	null
p1_well_no	Z-13	Z-13	null
p1_name_of_lease	Campbell Est "Z"	Campbell Est "Z"	null
p1_drilling_commenced	April 29, 1955	April 29, 1955	null
p1_party_plugging_well	R.W. Darden Dr'lg Co	R.W.Darden Drlg Co	null
p2_correspondence_name	R.L. Underwood	R.L.Underwood	null

Red: empty, or the two readings disagree (P and M differ only in spacing and punctuation).

P 59 / 77 fields filled

M 59 / 77 fields filled

O 1 / 77 fields filled

P: the reader agent on the page images. M: the reader agent on the MinerU parse and the images. O: the benchmark's own one-shot reader.

ExtractBench short: 2A-9-160298_109929; stored readings after their checks.

Step 4: checks that repair the structure of a reading

Check	Document	Observed change	Value F1
Rows copied into another list	Pueblo check register (medium)	**562 → 51 batch rows** Check 01033982 ($700.88) had also been filed as a batch with the same number and amount.	84.73 → 99.91
Identical records capped at the printed count	FTX creditor list, text layer (long)	**146 → 37 rows on page 59** Page 59 prints the line "NAME ON FILE" exactly 37 times.	84.47 → 90.77
Stacked lines folded into one record	Clinical adverse-event listing (long)	**1,499 → 668 records** Subject R1002: three printed lines became one event with its resolved date.	38.80 → 68.77
Template rows dropped	Texas W-14 well form (short)	**5 → 2 casing rows** Three rows carried only the printed row label (e.g. "Intermediate").	82.30 → 93.23

Examples are the cases each check was designed for. On a few short documents some repairs lowered the score slightly (e.g. duplicate removal on three election reports, −3.5 points in total).

Stored readings before and after each check, re-run with `docquorum/checks` and scored with ExtractBench's scorer. Each check uses only the schema and the document's own text; no model is called.

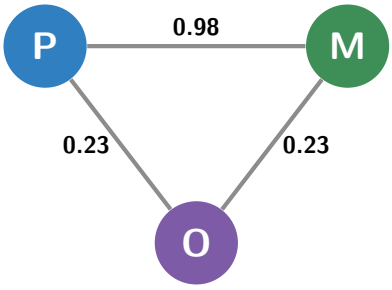
Step 4: checks that make values exactly what the page prints

Check	Document	Observed change	Value F1
Every schema field present	Caterpillar 330 spec sheet (short)	**missing keys → `null`** The reply omitted 7 optional fields in every specification row; the scorer counts an absent key as a miss.	61.48 → 95.74
A blank tick box is false	Texas P-4 form (short)	**6 boxes: `null` → `false`** The field description states "false if left blank".	91.67 → 96.67
Quotes as printed	Census statistical table (short)	**Bachelor's → Bachelor's** The page prints a typographic apostrophe; only the quote character is copied from the page.	60.12 → 62.58
Numbers as printed	Schedule K-1 (short, one-shot reading)	**33.333333 → 33.3333333** A rounded value is replaced by the single printed figure it restates (6 share fields).	85.42 → 97.92

Every replacement is a value that the document itself prints, or `null` where the schema admits null.

Stored readings before and after each check, re-run with `docquorum/checks` and scored with ExtractBench's scorer. Each check uses only the schema and the document's own text; no model is called.

Step 5: selection by quorum



Edge: agreement of two readings, measured with the benchmark's value scorer, recall-weighted (F_β, $\beta = 2$).

Reading	Mean agreement	
P image reading	0.604	**kept**
M parse reading	0.604	
O one-shot	0.218	

P and M tie; ties go to the reading listed first. The whole of P is returned; fields are never mixed across readings.

Agreement is not accuracy

Answer-key value F1 for this document (illustration only; never used by the pipeline): P 95.24, M 95.24, O 22.50. On the scanned county purchase order the quorum kept the one-shot reading (78.26), although the image reading scored higher (82.61).

ExtractBench short: 2A-9-160298_109929; agreement computed by `docquorum.quorum.pick` on the checked readings.

Step 6: one checked JSON, in the form requested

JSON Schema in, JSON out (1955 form, 77 fields)

```
{
  "p1_company":  "R. L. Underwood",
  "p1_address":  "420 Oil & Gas Bldg Wichita
Falls, Tex",
  "p1_county":  "Archer",
  "p1_well_no":  "Z-13",
  "p1_name_of_lease":  "Campbell Est \"Z\"",
  "p1_drilling_commenced":  "April 29, 1955",
  "p1_total_depth":  "1093'",
  ...  70 further fields
}
```

JSON template in, the same template filled



Template

```
{"menu":  [{"price":  "", "nm":  ""}],
 "total":  {"cashprice":  "",
            "total_price":  ""}}
```

Output (equal to the answer key)

```
{"menu":  [
   {"price":  "17500", "nm":  "J.STB PROMO"},
   {"price":  "46000", "nm":  "Y.B.BAT"},
   {"price":  "27500", "nm":  "Y.BASO PROM"},
 ],
 "total":  {"cashprice":  "91000",
            "total_price":  "91000"}}
```

Left: stored final output for 2A-9-160298_109929. Right: UNIKIE-BENCH Catering-Services test_1.jpg, quorum over six stored readings.

ExtractBench: complete results (1/3)

#	System	Type	Short	Medium	Long	Overall	$/page	s/doc
1	LlamaExtract Agentic Plus	Commercial service	96.94	94.45	90.18	95.91	0.081	152
1	Pulse (Effort)	Commercial service	96.46	95.01	93.51	95.91	0.105	165
3	**DocQuorum (ours)**	**Open model (27B)**	**96.34**	**94.11**	**90.41**	**95.43**	**0.029**[e]	**568**
4	LlamaExtract Agentic	Commercial service	96.04	92.11	91.83	94.77	0.033	80
5	Codex (GPT-6 Sol Evidence)	Coding agent	95.99	92.42	87.25	94.57	0.098	100
6	Codex (GPT-5.6 Sol Evidence)	Coding agent	96.04	90.19	82.73	93.77	0.216	104
7	Codex (GPT-5.5)	Coding agent	95.68	91.15	78.88	93.57	0.278	76
8	Codex (GPT-5.5 Evidence)	Coding agent	95.59	88.72	87.97	93.35	0.336	101
9	Pulse (Non-effort)	Commercial service	93.17	92.22	89.07	92.70	0.030	113
10	LlamaExtract Cost-Effective	Commercial service	94.03	89.51	83.73	92.27	0.011	120
11	Codex (GPT-5.6 Terra Evidence)	Coding agent	95.60	86.69	77.36	92.26	0.095	78
12	OpenAI GPT-6 Astra	Commercial model	97.22	90.56	31.70	91.91	0.111	57
13	Codex (GPT-5.6 Luna Evidence)	Coding agent	93.50	87.43	80.05	91.16	0.012	126
14	Codex (GPT-6 Luna Evidence)	Coding agent	93.09	87.71	80.88	91.01	0.010	219
15	Reducto Deep Extract	Commercial service	94.20	80.47	92.01	90.44	0.344	250
16	Qwen3.8 Flash Next	Open model	94.82	87.81	37.74	89.88	self-host	188

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems) and DocQuorum. Value F1 (%) by the benchmark's scorer over 370 PDFs (252 short, 98 medium, 20 long); Overall is the mean over all documents. Equal scores share a rank. $/page and s/doc: the leaderboard's cost per page and latency per document (averaged over the 370 documents); self-host: an open model the leaderboard ran itself and did not price; n/p: not published. e: DocQuorum's tokens measured on a random 20, priced at the Qwen3.8-27B list price (Alibaba Cloud, $0.50 / $3.00 per million input / output tokens) and weighted by split as the leaderboard's $/page is; timed one document at a time on a GPU shared with other jobs.

ExtractBench: complete results (2/3)

#	System	Type	Short	Medium	Long	Overall	$/page	s/doc
17	Qwen3.8 27B	Open model	94.68	87.54	38.45	89.75	self-host	274
18	Extend (Max Context)	Commercial service	92.32	78.75	90.46	88.62	0.100	113
19	Qwen3.6 35B	Open model	92.86	86.37	36.90	88.11	self-host	97
20	Qwen3.5 35B	Open model	93.52	85.27	31.73	88.00	self-host	86
21	DeepSeek V4.1 Flash	Commercial model	94.44	81.49	22.23	87.11	0.003	16
22	Claude Code (Opus 4.8)	Coding agent	90.08	79.21	88.07	87.09	0.162	107
23	Claude Code (Opus 4.8 Evidence)	Coding agent	90.54	79.17	77.94	86.84	0.632	306
24	Qwen3.5 9B	Open model	92.31	81.12	32.22	86.10	self-host	116
25	Claude Code (Opus 5.5 Evidence)	Coding agent	89.63	77.96	80.83	86.06	0.121	88
26	Extend Extract	Commercial service	89.39	84.14	47.21	85.72	0.068	120
27	Datalab (Accurate + Balanced)	Commercial service	89.40	85.08	42.04	85.70	0.035	294
28	Reducto Extract	Commercial service	92.97	69.59	30.65	83.41	0.055	64
29	Kimi K3	Commercial model	94.64	69.64	4.99	83.17	0.068	159
30	Qwen3.5 4B	Open model	89.06	76.64	27.31	82.43	self-host	102
31	NuExtract3	Open model	86.41	77.01	32.71	81.02	self-host	67
32	GLM-5.3 Flash	Commercial model	96.30	51.56	27.83	80.75	0.003	87

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ExtractBench: complete results (3/3)

#	System	Type	Short	Medium	Long	Overall	$/page	s/doc
33	Google Gemini 3.8 Flash	Commercial model	88.14	72.23	28.70	80.71	0.004	16
34	Google Gemini 3.5 Flash	Commercial model	87.87	69.76	27.90	79.84	0.010	20
35	Lift Datalab 9B	Open model	87.64	65.69	24.78	78.43	self-host	89
36	OpenAI GPT-5.4 Nano	Commercial model	77.43	76.37	35.81	74.90	0.002	43
37	Gemma4 26B	Open model	77.70	59.61	28.88	70.27	self-host	653
38	Gemma4 E4B	Open model	81.10	48.93	26.82	69.64	self-host	50
39	InternVL3.5 14B	Open model	80.70	39.04	19.41	66.35	self-host	n/p
40	InternVL3.5 8B	Open model	79.26	36.62	17.57	64.63	self-host	n/p
41	Qwen3.5 2B	Open model	71.22	50.93	25.93	63.40	self-host	214
42	Kimi-VL A3B	Open model	69.86	34.61	18.94	57.77	self-host	71
43	Gemma4 E2B	Open model	63.71	28.96	14.58	51.85	self-host	48
44	NuExtract 2.0 8B	Open model	62.92	23.50	6.39	49.42	self-host	n/p
45	Qwen3.5 0.8B	Open model	57.44	26.98	6.65	46.63	self-host	191
46	MiniCPM-V 4.5	Open model	57.39	25.30	1.13	45.85	self-host	n/p
47	Gemma4 12B	Open model	53.62	30.82	13.16	45.39	self-host	241
48	GLM-4.6V Flash	Open model	47.28	44.52	22.17	45.19	self-host	226

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems) and DocQuorum. Value F1 (%) by the benchmark's scorer over 370 PDFs (252 short, 98 medium, 20 long); Overall is the mean over all documents. Equal scores share a rank. $/page and s/doc: the leaderboard's cost per page and latency per document (averaged over the 370 documents); self-host: an open model the leaderboard ran itself and did not price; n/p: not published. e: DocQuorum's tokens measured on a random 20, priced at the Qwen3.8-27B list price (Alibaba Cloud, $0.50 / $3.00 per million input / output tokens) and weighted by split as the leaderboard's $/page is; timed one document at a time on a GPU shared with other jobs.

UNIKIE-BENCH: complete results

#	System		Ret.	Cat.	Com.	Acc.	Post.	Adm.	Edu.	Adv.	Tax	Med.	Nut.	Avg.
1	**DocQuorum (ours)**	O	**90.63**	**88.13**	**53.51**	**79.29**	**85.65**	**87.00**	**85.67**	**86.59**	**82.16**	**78.33**	**92.22**	**82.65**
2	Gemini-3-Pro	C	89.27	83.54	53.87	81.43	90.74	84.32	80.71	90.54	81.54	77.60	92.47	82.37
3	Qwen3-VL-Plus	C	89.12	86.45	50.57	77.86	83.55	78.91	87.44	83.14	78.93	76.46	89.72	80.20
4	Qwen3-VL-8B	O	88.26	86.23	51.14	71.07	83.80	78.90	87.39	81.23	78.74	77.19	86.36	79.12
5	Qwen-VL-Max	C	86.67	84.68	49.33	76.07	84.85	79.96	82.46	82.76	77.94	75.62	75.14	77.77
6	GLM-4.1V-9B	O	62.32	77.33	46.67	76.43	76.46	78.75	65.83	81.99	70.68	72.08	60.47	69.91
7	GPT-4o	C	75.79	79.33	49.67	67.86	82.95	64.26	47.77	73.56	75.33	67.57	76.51	69.15
8	MiMo-VL-7B-RL	O	77.45	77.66	49.01	67.50	76.70	76.81	76.65	66.28	74.86	68.70	46.10	68.88
9	InternVL3.5-8B	O	77.23	72.23	41.64	67.14	73.90	72.10	63.08	73.08	72.81	65.94	65.53	67.70
10	GPT-5	C	71.90	71.33	45.94	69.29	80.95	62.79	50.26	75.48	72.05	67.71	75.83	67.59
11	MiniCPM-V4.5-8B	O	68.37	74.91	45.91	67.76	83.60	71.89	63.21	79.31	69.57	70.00	44.18	67.15
12	Claude-Sonnet-4.5	C	69.24	73.65	45.92	69.29	79.45	69.20	39.46	85.06	67.79	63.96	67.24	66.39
13	Kimi-VL-A3B	O	86.49	63.12	42.87	61.79	83.40	68.72	80.37	72.41	77.80	64.93	25.72	66.14
14	Gemma-3-12B	O	54.32	68.73	30.68	53.93	72.30	37.14	17.56	36.78	56.81	48.54	52.20	48.09
15	Ministral-3-8B	O	6.47	57.01	26.24	43.39	75.80	32.37	29.54	19.53	65.85	51.46	6.17	37.62
16	SmolVLM2-2.2B	O	17.80	57.22	28.52	18.92	34.98	8.63	3.87	27.16	34.97	18.00	10.51	23.69

Ret. Retail; Cat. Catering services; Com. Commercial; Acc. Accommodation; Post. Postal label; Adm. Administrative; Edu. Education; Adv. Advertisement; Tax: Tax-compliant; Med. Medical services; Nut. Nutrition label; Avg. macro average. C: closed model; O: open model. Underlined: best in the column.

UNIKIE-BENCH (ACL 2026), Table 4: 15 published systems; F1 (%) by the benchmark's evaluator on 4,472 images, every input capped at 1,605,632 pixels. DocQuorum: quorum over six readings from three open models; small images enlarged before reading.

VAREX: complete results

#	System	Size	EM	ANLS	Flat	Nested	Table	Perfect
1	**DocQuorum (ours)**	**27B**	**98.2**	**98.8**	**98.4**	**98.5**	**97.4**	**86.2**
2	Gemini 2.5 Pro	API	98.0	98.5	97.3	98.2	97.7	82.8
3	Gemini 2.5 Flash	API	97.3	97.9	96.9	97.7	96.5	81.0
4	Qwen3-VL (8B)	8B	96.6	97.6	96.6	97.3	95.0	78.2
5	Llama 4 Maverick (17B×128E)	17B×128E	95.6	97.1	95.1	96.1	94.4	68.7
6	Ministral (14B)	14B	94.8	96.5	94.8	95.1	94.0	66.0
6	GPT-4o	API	94.8	97.1	95.1	95.3	93.5	64.3
8	Llama 4 Scout (17B×16E)	17B×16E	94.3	95.8	95.2	94.3	94.1	63.4
9	granite-vision-4.1-4b (4B)	4B	94.2	96.4	95.2	94.9	92.3	65.2
10	NuExtract 2.0 (2B)	2B	90.8	92.1	93.1	91.9	87.3	52.7
11	InternVL3.5 (2B)	2B	85.6	89.1	89.6	86.1	82.7	42.4
12	granite-4.0-3b-vision (3B)	3B	85.5	89.1	91.5	85.2	84.0	41.1
13	Qwen 2.5-VL (3B)	3B	82.5	85.4	79.9	83.1	81.9	43.7
14	Gemma 3n (E2B)	E2B	71.0	78.1	76.7	71.0	68.8	19.0
15	MiniCPM-V4 (4B)	4B	67.9	69.8	84.8	66.2	65.5	23.6
16	Gemma 3 (4B)	4B	65.3	75.0	71.7	65.0	63.6	13.7
17	h2oVL Miss. (2B)	2B	61.3	64.9	67.2	61.3	58.9	11.6
18	Qwen3-VL (2B)	2B	34.2	34.7	93.3	26.8	29.0	29.3
18	h2oVL Miss. (800M)	800M	34.2	36.1	35.8	33.2	36.0	5.9
20	InternVL3.5 (1B)	1B	28.2	29.8	79.0	22.4	22.2	11.2
21	Qwen2-VL (2B)	2B	9.7	10.0	16.4	9.0	8.9	4.3

EM: exact match (normalized) over 21,084 fields; ANLS: normalized ANLS; Flat, Nested, Table: exact match by form type (299, 1,146, 332 forms); Perfect: forms with every field right. Underlined: best in the column.

VAREX-Bench image-modality leaderboard (20 models, zero-shot, one decimal) and DocQuorum (final pipeline). Ranks compare at one decimal.

LongArray-Extract: complete results

#	System	Accuracy	Completed
1	Extend MAX	99.2	45/45
2	**DocQuorum (ours)**	**98.1**	**45/45**
3	Reducto Deep Extract	97.4	45/45
4	Claude Opus 4.7	83.1	45/45
5	Reducto Standard	80.9	45/45
6	Pulse Effort	68.8	45/45
7	Pulse Auto	64.5	45/45
8	Gemini 3.1 Pro	47.3	45/45
9	LlamaParse Standard	47.2	26/45
10	LlamaParse Agentic	34.7	35/45
11	GPT-5.5	31.6	45/45
12	Gemini 3.5 Flash	31.1	45/45

Extend AI's LongArray-Extract results page (11 systems; one decimal) and DocQuorum: macro document accuracy (%) by the publisher's reference scorer over 45 long documents (bank statements, clinical listings, indictments; 29,328 rows); a failed document counts as zero. Ranks compare at one decimal.

UNIKIE-BENCH open category: complete results

#	System		Ret. zh	Form zh	Inv. zh	Con. zh	Ret. en	Form en	Inv. en	Con. en	Avg.
1	Gemini-3-Pro	C	78.50	79.17	78.63	75.96	93.57	80.41	87.06	79.90	81.65
2	**DocQuorum (ours)**	**O**	**67.67**	**69.80**	**69.72**	**74.25**	**83.10**	**67.65**	**68.22**	**78.22**	**72.33**
3	Qwen3-VL-Plus	C	71.14	69.87	69.59	70.95	68.27	66.82	73.45	74.81	70.61
4	Qwen-VL-Max	C	69.94	69.75	66.62	69.68	70.76	65.52	70.80	76.57	69.95
5	Qwen3-VL-8B	O	68.93	64.70	63.38	65.59	71.40	61.46	67.34	76.08	67.36
6	MiMo-VL-7B-RL	O	57.63	56.60	51.93	65.90	71.34	52.83	57.64	70.55	60.55
7	Kimi-VL-A3B	O	62.98	52.85	50.72	54.94	73.91	55.96	52.67	64.64	58.58
8	GLM-4.1V-9B	O	55.19	56.63	55.52	54.68	63.83	54.66	59.31	68.70	58.56
9	MiniCPM-V4.5-8B	O	52.33	42.69	44.54	39.46	72.41	47.46	56.44	65.49	52.60
10	InternVL3.5-8B	O	54.26	37.83	40.97	39.43	79.46	43.64	49.41	50.06	49.38
11	GPT-5	C	39.61	31.97	37.64	17.47	80.64	50.09	57.50	64.24	47.39
12	GPT-4o	C	39.66	30.82	35.31	16.67	78.51	48.09	57.08	67.35	46.69
13	Ministral-3-8B	O	25.21	35.80	24.42	26.18	58.54	49.01	55.66	57.82	41.58
14	Claude-Sonnet-4.5	C	37.59	31.20	29.12	20.88	41.64	49.22	44.18	63.87	39.71
15	Gemma-3-12B	O	25.71	14.17	17.36	9.42	76.24	38.32	30.23	35.77	30.90
16	SmolVLM2-2.2B	O	5.74	1.51	3.05	2.15	36.49	11.64	19.73	22.36	12.83

Ret. receipts; Inv. invoices; Con. contracts; zh Chinese, en English; Avg. macro average. C: closed model; O: open model. Underlined: best in the column.

UNIKIE-BENCH (ACL 2026), open-category table: 15 published systems; F1 (%) by the benchmark's evaluator on 1,661 documents, every input capped at 1,605,632 pixels. DocQuorum: quorum over three readings from Qwen3.8-27B, then zoom re-reading.

DocuBench: complete results

#	System	Accuracy	$/doc	s/doc
1	DocuPipe (high effort)	97.02	0.54	196
2	DocuPipe (standard effort)	96.14	0.11	73
3	Claude Sonnet 5 (direct LLM)	91.73	0.29	91
4	**DocQuorum (ours)**	**91.54**	**0.21**[e]	**261**
5	Reducto Deep Extract	89.38	0.70	204
6	Reducto (standard)	81.11	0.36	66
7	Extend	80.28	0.40	49
8	GPT-5.5 (direct LLM)	76.48	0.18[c]	71
9	Gemini 3.5 Flash (direct LLM)	72.98	0.12[c]	54
10	Unstructured	67.67	0.22[c]	630

DocuBench v0.2.0 leaderboard (9 systems) and DocQuorum (final pipeline): macro field accuracy (%) by the benchmark's own scorer over 72 documents in 10 file types (PDF, scans, photos, spreadsheets, Word, HTML, XML, CSV, text). The final pipeline's fixes were found on this benchmark. $/doc and s/doc: means over each system's 72 result files of its recorded cost and time; c: computed from its recorded tokens at list price (GPT-5.5 $5 / $30, Gemini 3.5 Flash $1.50 / $9 per million input / output tokens) or, for Unstructured, its pages at DocuBench's $0.03 per page. e: DocQuorum's tokens measured on a random 20 documents, priced at the Qwen3.8-27B list price (Alibaba Cloud, $0.50 / $3.00); it runs on its own GPU, one document at a time on a shared H100 for s/doc.

Results on six benchmarks

Benchmark	DocQuorum	Place	Comparison	Calls	$	s
VAREX (1,777 forms)	**98.23**	**1st of 21**	next: Gemini 2.5 Pro 98.0	3.7	0.021	25
UNIKIE-BENCH (4,472 images)	**82.65**	**1st of 16**	next: Gemini-3-Pro 82.37	6.3	0.010	51
LongArray-Extract (45 documents)	**98.05**	2nd of 12	1st: Extend MAX 99.2	100.3	0.676	1,466
UNIKIE-BENCH open (1,661 documents)	**72.33**	2nd of 16	1st: Gemini-3-Pro 81.65	4.6	0.018	121
ExtractBench (370 PDFs) (development set)	**95.43**	3rd of 48	1st: LlamaExtract Agentic Plus 95.91	41.7	0.407	568
DocuBench (72 documents)	**91.54**	4th of 10	1st: DocuPipe (high effort) 97.02	11.1	0.212	261

Each score is the benchmark's own scorer on the full test set, with open-weight models (Qwen3.8-27B and MinerU2.5) on one GPU. Places count every published system. Calls, dollars ($) and seconds (s): means per document, measured on a random sample one document at a time on one shared H100 (results/costs/); $ prices the measured tokens at the Qwen3.8-27B list price, since the pipeline itself pays no price per call.

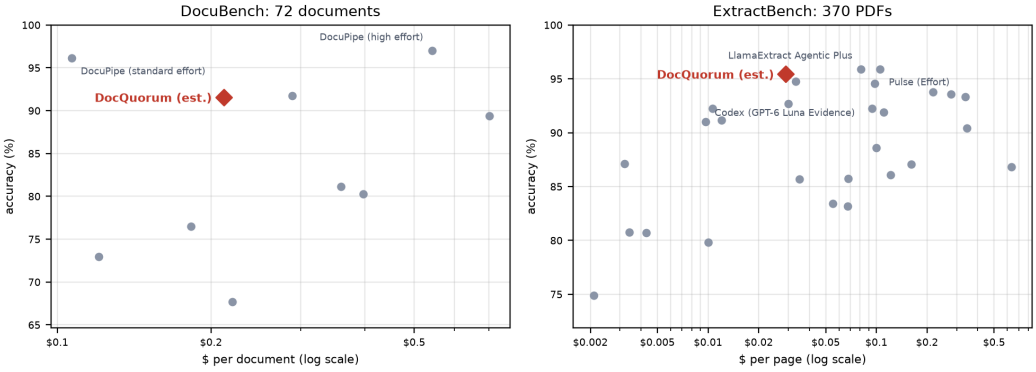
docs/results/: one complete table per benchmark (TABLES.md). No result was submitted to a leaderboard.

Where DocQuorum stands out

Benchmark	Place, score	Ahead of	Per document (measured)
VAREX	1st, 98.23	Gemini 2.5 Pro 98.0 (closed); best open, Qwen3-VL (8B) 96.6	3.7 calls, $0.021/form
UNIKIE-BENCH	1st, 82.65	Gemini-3-Pro 82.37 (closed); best open, Qwen3-VL-8B 79.12	6.3 calls, $0.010/image
ExtractBench	3rd, 95.43	the same Qwen3.8-27B called directly 89.75; every coding agent (best Codex (GPT-6 Sol Evidence) 94.57, $0.098/page)	$0.029 per page
LongArray-Extract	2nd, 98.05	Reducto Deep Extract 97.4; Claude Opus 4.7 83.1; GPT-5.5 31.6	100.3 calls, $0.676/document
UNIKIE-BENCH open	2nd, 72.33	best open, Qwen3-VL-8B 67.36; Qwen3-VL-Plus 70.61 (closed)	4.6 calls, $0.018/document
DocuBench	4th, 91.54	Reducto Deep Extract 89.38 at $0.70; Extend 80.28 at $0.40; GPT-5.5 76.48 at $0.18	$0.21 mean; 13 of 20 in one call (median $0.011)

Open models only: Qwen3.8-27B and the 1.2B MinerU2.5 parser on one GPU. $ prices the measured tokens at the Qwen3.8-27B list price; published costs are each system's own (DocuBench result files, ExtractBench leaderboard). UNIKIE-BENCH: two of the six readings come from Qwen3-VL-32B and -8B, counted as calls but not priced.

Accuracy against cost



Grey: every published system with a cost; red: DocQuorum, its measured tokens priced at the Qwen3.8-27B list price ($0.50 / $3.00 per million input / output tokens), mean of a random sample (20 DocuBench documents, 20 ExtractBench documents). Published costs: DocuBench result files (some computed from their tokens) and the ExtractBench leaderboard.

Why a document costs little, and when it costs more

Why it costs little

One reading when the file has text. On DocuBench, 13 of 20 sampled documents took one model call, $0.009 each.

More readings only for scans; zoom re-reads only disputed fields a new vote could change. A VAREX form takes 3.7 calls (three readings plus zoom).

An open 27B model: $3 per million output tokens at list price, a tenth of GPT-5.5 ($30) and a fifth of Claude Sonnet 5 ($15); on one GPU, no price per call at all.

Accuracy from the pipeline, not a bigger model. Qwen3.8-27B called directly scores 89.75 on ExtractBench; DocQuorum with it, 95.43.

When it costs more

Scanned documents get three readings (at the same time) and a quorum: 7 sampled DocuBench documents, 30 calls and $0.59 each on average.

Long scans cost most: they are read in chunks, once per reading. The costliest sample, FnscKwQQ (16 pages), took 44 calls, $1.04 and 18 minutes.

Long documents are read in chunks, so they cost by the page: LongArray's sampled documents (48 pages on average) took 100 calls and $0.68 each, $0.014 a page; ExtractBench, $0.029 a page.

results/costs/: every measured document (calls, tokens, seconds); prices in docs/results/pricing.json.

v1.4: long scans read faster

A looping reply is stopped by the model server once one stretch of tokens repeats 40 times in a row, not at its output limit; it is then read again as before.

A scan's readings run at the same time, not one after another; the quorum sees them in the same order.

The same documents, measured alone both times:

DocuBench (20): 12,564 s → **5,225 s**, 2.4× faster
VAREX (30 forms): 1,178 s → **750 s**, 1.6× faster

Accuracy kept: DocuBench, all 72 read again, 91.54 (v1.3: 90.80); 29 of 30 sampled VAREX forms unchanged.

Slowest five	pages	minutes		output tokens	
		v1.3	v1.4	v1.3	v1.4
f9nrfUNC	10	61	11	224k	81k
EWUUeFYp	28	39	16	154k	69k
kcnZ5EWa	3	35	16	89k	69k
FnscKwQQ	16	30	18	113k	112k
sHMajYzm	16	27	16	111k	112k

DocuBench's five slowest sampled documents. results/costs/v1.3 and results/costs: every measured document, before and after.

Zoom re-reading: a new step after the quorum

After the quorum keeps one reading, each field on which the readings disagree is found on the page and read once more from a sharp crop of that region.

This new reading sees none of the earlier answers. It changes the kept value only if it matches a value some reading already proposed, and it counts as two votes: it can overturn a lead of one reading, not a clear majority.

The vote rule was added after the third test, where five agreeing readings were overturned too often.

Three tests

VAREX: 98.10 → **98.30**
60 forms better, 25 worse

UNIKIE open: 71.07 → **72.33**
291 documents better, 138 worse

UNIKIE (six readings): 82.62 → 81.62;
82.65 with the vote rule

results/README.md: method, statistics, and every changed value checked against the answer key after the runs.

Summary

Results

1st on VAREX and UNIKIE-BENCH; 2nd on LongArray-Extract and UNIKIE open; 3rd on ExtractBench.

A new step, zoom re-reading, improved VAREX and UNIKIE open; with its vote rule it holds UNIKIE level.

Open-weight models on one GPU.

Method

Routing by the available evidence; an agentic reader that plans, reads and re-reads; layout readers for column flow and ruled tables; deterministic checks against the schema and the page; selection by quorum.

Limitations

ExtractBench served as the development set. Agreement among readings does not guarantee accuracy. Checks are triggered by schema wording and page text; a few repairs lower the score on some documents.